Daniel Martín Guerrero

BSc in Physics (2025–2029)
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Physics · Optics & Photonics · Astrophysics · Quantum Mechanics · Science Communication

EDUCATION

Universidad de Salamanca (USAL) 2025 – Present
BSc in Physics – second year, expected June 2029 Salamanca, Spain

- Awarded **Honours** (*Matrícula de Honor*) in *Physics Laboratory*.
- Holder of the **University Access Award** (*Premio de Acceso a la Universidad*), a full first-year tuition waiver.

Upper Secondary Education (Bachillerato) – Science and Technology track 2023 – 2025
IES Federico García Bernalt Salamanca, Spain

- Graduated with **Honours** (*Matrícula de Honor*); overall average **9.88/10**.
- University entrance examination (PAU): **13.378/14**.

RELEVANT COURSEWORK

Linear Algebra & Geometry Matrix and tensor calculus; vector spaces; linear, bilinear and sesquilinear maps (endomorphisms, metrics, polarities, quadratic forms); dual spaces and tensors; affine and Euclidean geometry; foundations of differential geometry.

Mathematical Analysis Differential and integral calculus in one and several variables; vector calculus; sequences and series; Taylor expansions and analysis of functions; complex variable.

Differential Equations Modelling of physical and natural systems: Malthusian and logistic population dynamics, Gompertz learning curves, SIS epidemic models, Lotka–Volterra systems, RLC circuits via the Laplace transform and signal analysis via the Fourier transform; introduction to topology, stochastic processes and measure theory.

Physics Classical mechanics, oscillations and waves (fundamentals of optics), electromagnetism, thermodynamics and computational techniques in physics.

Experimental Physics Data treatment and uncertainty quantification in the laboratory and through simulation: statistical estimation, covariance and propagation of variances, weighted means, least-squares fitting with χ^2 model assessment and coverage intervals (Welch–Satterthwaite).

INDEPENDENT STUDY & PROJECTS

Foundations of Quantum Mechanics and Quantum Computing 2026 – Present
Independent study – author of structured LaTeX lecture notes

- Authored *Lesson 1: Hilbert Spaces and the Dirac Formalism – Mathematical Foundations and the Born Postulate*, covering sesquilinear forms and inner products, Cauchy completeness and Hilbert spaces, the space $L^2(\mathbb{R})$ and square integrability, the Cauchy–Schwarz inequality, bras and kets, Hermitian adjoints, observables and spectral decomposition, and the Born rule and interference of probability amplitudes.
- Studying the fundamentals of quantum computation: qubits, observables and quantum logic gates (e.g. Hadamard).

AI-Assisted Study, Simulation and Modelling Workflow 2025 – Present
Personal project – Mathematics & Physics

- Use frontier and open-source language models to produce structured study notes and to build simulations and models of physical situations, from parametrised manifolds to linear least-squares fits, as well as to formalise and develop new ideas.
- Use NotebookLM as a learning aid for problem solving and for the clear visualisation of complex concepts.

Tools: Claude Opus 5.5, Gemini, open-source language models, NotebookLM, Python, LaTeX.

SCIENCE COMMUNICATION & OUTREACH

***Eureka* – Science Outreach Programme, Radio USAL** 2026 – Present
Scriptwriter & Presenter; creator of the segment Cosmosofía Salamanca, Spain

- Write and present science-outreach content for the official radio station of the University of Salamanca, within the *Radioacción Cósmica* slot of the Supernova astronomy association.
- Created *Cosmosofía*, a segment bridging astrophysics and the philosophy of science. Episode I, *"A Sea of Four Hundred Billion Suns"*, explores our cosmic coordinates, stellar nucleosynthesis and Cecilia Payne's discovery, science as an open conversation, and Sagan's *Pale Blue Dot*.

Physics & Mathematics Communication on TikTok – @_.danielmg Ongoing

Content creator

- Produce AI-assisted videos, from general outreach to specialised topics, ranging from astrophysics, electromagnetism and thermodynamics to Sylvester's law of inertia, linear differential forms and the generalised Stokes theorem.

European Researchers' Night 2026 – UCC+i, Universidad de Salamanca 25 Sep. 2026

Outreach activity "Houston, tenemos un telescopio" (Houston, we have a telescope) Salamanca, Spain

- Delivered a three-hour public astronomy activity within the Scientific Culture and Innovation Unit (UCC+i) programme 2025–2026 (certified).

TALKS & PUBLIC ENGAGEMENT

Invited Speaker & Round-Table Panellist – XXI Philosophical Olympiad of Castilla y León 27 Feb. 2026

Universidad de Salamanca – theme: Madness: Don Quixote in Today's World Salamanca, Spain

- Delivered the talk *"El naufragio de la luz. Disertación filosófica acerca de la lucidez del delirio. Don Quijote, Myshkin y la estética de la bondad"* (*The Shipwreck of Light: a philosophical dissertation on the lucidity of delirium – Don Quixote, Myshkin and the aesthetics of goodness*); certified by the Philosophical Olympiad Association of Castilla y León.

Panellist – Presentation of *Salvar a Sócrates* (Eduardo Infante, Ariel) 5 Jun. 2026

Librería Víctor Jara Salamanca, Spain

- Public conversation with the philosopher and author Eduardo Infante on his novel for learning to philosophise, alongside fellow panellists.

HONOURS & AWARDS

- **University Access Award**, Universidad de Salamanca – granted to the 100 highest scores in the general phase of the 2024/25 university entrance examination 2025
- **PAU score of 13.378/14**, Spanish university entrance examination 2025
- **Honours** (*Matrícula de Honor*), Bachillerato – average 9.88/10 2025
- **Honours** (*Matrícula de Honor*) in *Physics Laboratory*, BSc in Physics, USAL 2026
- **Finalist**, Philosophical Olympiad of Castilla y León (OFCyL), philosophical dissertation category (Bachillerato) – two consecutive editions (XIX and XX) 2024, 2025
- **Participant**, PLANCKS 2027 – international theoretical physics competition Upcoming

MEMBERSHIPS & AFFILIATIONS

- **Optica** (Advancing Optics and Photonics Worldwide) – Individual Member Through Sep. 2027
- **OSAL Student Chapter** – student association for optics outreach, Universidad de Salamanca Current
- **Supernova** University Astronomy Association – outreach events, observation nights and retreats; radio scriptwriter and presenter 2025 – Present
- **Delegación Charra**, Student Group (GdeE) of the Royal Spanish Physics Society (RSEF) Current

TECHNICAL SKILLS

Programming	Python, MATLAB, Arduino (C/C++-based language built on the Wiring framework), HTML (basic).
Electronics & Lab	Breadboard circuit prototyping with Arduino and LabVIEW; hands-on experimental work, measurement and instrumentation.
Scientific Tools	Mathematica, PHYWE measure, LabVIEW, Arduino IDE, SAOImage DS9 (astronomical image analysis), LaTeX (advanced).
AI Tools	Claude Opus 5.5, Gemini, open-source language models, NotebookLM.
Languages	Spanish (native); English (advanced – trained up to C1 level; certified through the PAU English examination).

INTERESTS

Physics	Astrophysics; optics and photonics (fundamental laser science, integrated photonics); particle physics; quantum mechanics and quantum computing.
Philosophy & Film	Philosophy of science and aesthetics; philosophy through cinema – Tarkovsky, Lynch, Bergman, Kurosawa and Masaki Kobayashi. Film reviewer on Letterboxd (danielmtg).